Hugo Marquerie 11/05/2025

Teorema de Liouville

Teorema 1 (de Liouville). *Toda función entera y acotada es constante.*

***Demostración*:** Observamos que es suficiente pedir que f esté acotada en un conjunto del tipo $\{z \in \mathbb{C} : |z| > R\}$, donde $R > 0$.

Aplicamos la fórmula integral de Cauchy y que $\forall z \in \{z \in \mathbb{C} : |z| > R\} : |f(z)| \leqslant M$:

$$|f'(z)| \leqslant \frac{1}{2\pi} \int_{C_R^+} \frac{M}{R^2} \,|\mathrm{d}w| = \frac{1}{2\pi} \cdot \frac{M}{R^2} \cdot \operatorname{long}(C_R^+) = \frac{M}{R} \xrightarrow{R \to \infty} 0 \implies f' \equiv 0.$$

Por tanto, f es constante. ■

Referenciado en

- `teo-fundamental-algebra`

Temporary page!

LaTeX was unable to guess the total number of pages correctly. As there was some unprocessed data that should have been added to the final page this extra page has been added to receive it.

If you rerun the document (without altering it) this surplus page will go away, because LaTeX now knows how many pages to expect for this document.